EFY Note
The source code of this project is available for free download at *source.efymag.com*

the channels, such as temperature, humidity and voltage. Channel ID in this example is 279012, as shown in Fig. 2. This channel ID will be different in your case.

Eight data per channel. Each channel can take eight data fields from different devices. That means, using ThingSpeak API you can upload eight data per channel, which will eventually be gathered, logged and put into a trend data by ThingSpeak, as shown in Fig. 3.

Installation of ESP8266 board

First, install Arduino 1.6.4, or latest version, from Arduino website. Start Arduino and install ESP8266 support file from Arduino IDE's board manager.

Next, open Preferences and enter the following URL under Additional Board Manager URLs (Fig. 4):

http://arduino.esp8266.com/package_esp8266com_index.json

You can add multiple URLs, separating these with commas.

Open Boards Manager from Tools > Board menu and install esp8266 platform. Do not forget to select your ESP8266 board from Tools > Board menu after installation. The board used here is WeMos D1 R2. Go to Tools > Boards and find WeMos D1 R2 board from the drop-down list.

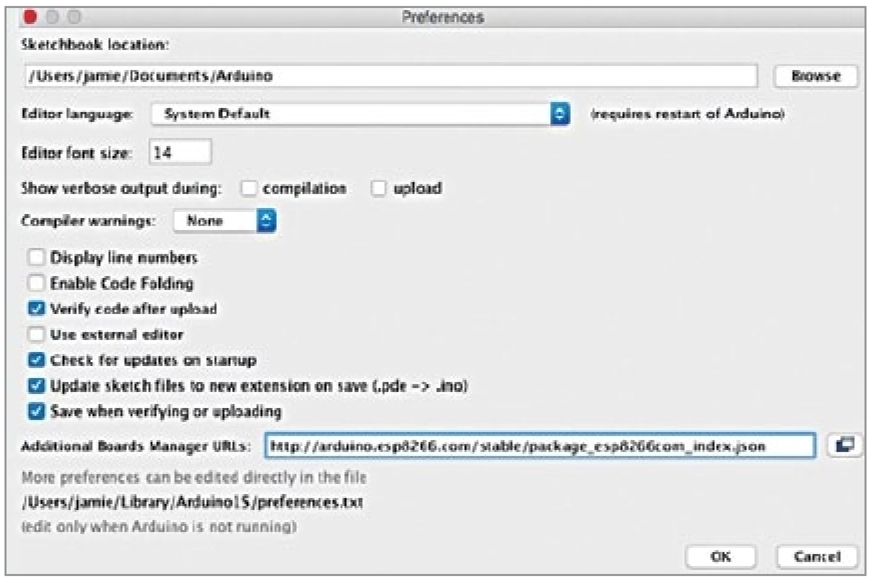

Fig. 4: Arduino JSON configuration

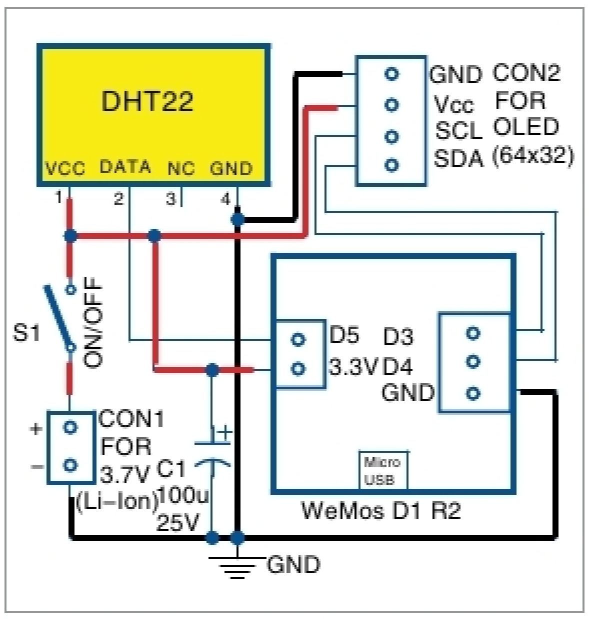

Fig. 5: Circuit diagram of the data logger

Circuit diagram

The circuit diagram of the IoT-based data logger is shown in Fig. 5. It consists of DHT22 sensor, WeMos D1 R1 board, OLED display and 3.7V lithium-ion cell (CON1). A simple DHT22 sensor is used here to upload temperature and humidity data to *ThingSpeak.com* channel. Six more data fields can be added, for which you will need to add six more sensors to the circuit.

For viewing these data, an optional OLED display (64x32) can be attached to the project at connector CON2. If you want to run on battery, remove the OLED for longer operation, as it consumes about 40mA current. The author's prototype is shown in Fig. 6. (OLED display section was not tested at EFY Lab.)

Software

The source code/sketch (IoT_datalogger.ino) is used for transferring the data to *ThingSpeak.com* channel. The code is broadly divided into different parts. The first part includes relevant headers to create instances of Wi-Fi, DHT and *ThingSpeak.com*

Next, the program captures temperature and humidity data, and after a delay of 15 seconds each transfers the data to *ThingSpeak.com*. Fifteen seconds is the minimum gap required by *ThingSpeak.com* for filling in successive channels.

Once WeMoS board is installed in Arduino IDE, compile the sketch and upload it to WeMos board. If the circuit connections and channel setup on ThingSpeak are correct, you will see the data as shown in Fig. 3.

If you declare a channel public, it will have a URL. Anyone can click that URL to view the channel; else, you have to log on to that channel to see the output. EFY

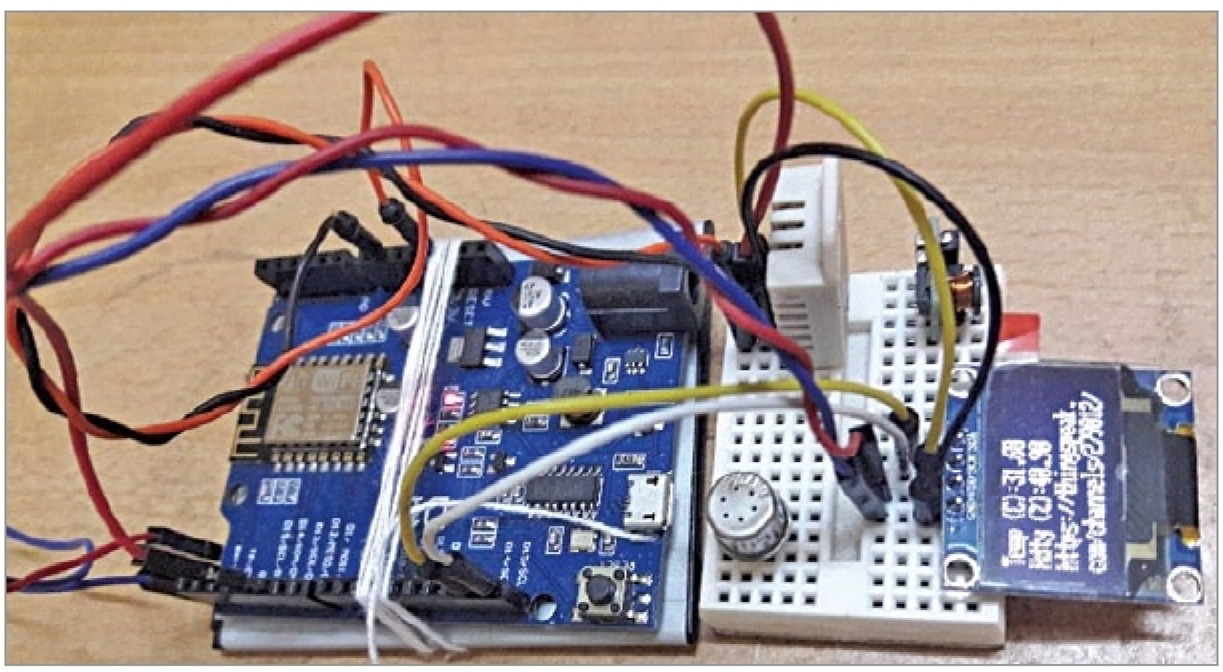

Fig. 6: Author's prototype with OLED display

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